

Main Conclusions

- Results on graph models and empirical networks show a large decrease in anonymity when accounting for knowledge about 2-neighborhoods
- When assessing node anonymity, it is important to account for more than a node's ego-network

Problem Statement

How anonymous are you in a social network, based on your structural position?

Measuring Anonymity

We measure anonymity by partitioning the nodes into **equivalence classes**. If a node is in an equivalence class of size k , there are k candidate nodes and the node is **k -anonymous**. The larger the value of k , the more anonymous the node is in the network. When there is only one candidate, the node is **unique**.

Equivalence

We propose to define equivalence based on a specific level of **attacker knowledge**. Our measure d - k -anonymity assumes perfect knowledge of the d -neighborhood surrounding a node. We compute d - k -anonymity by building upon state-of-the-art subgraph isomorphism tool Nauty [1] for isomorphisms as part of [2, 3].

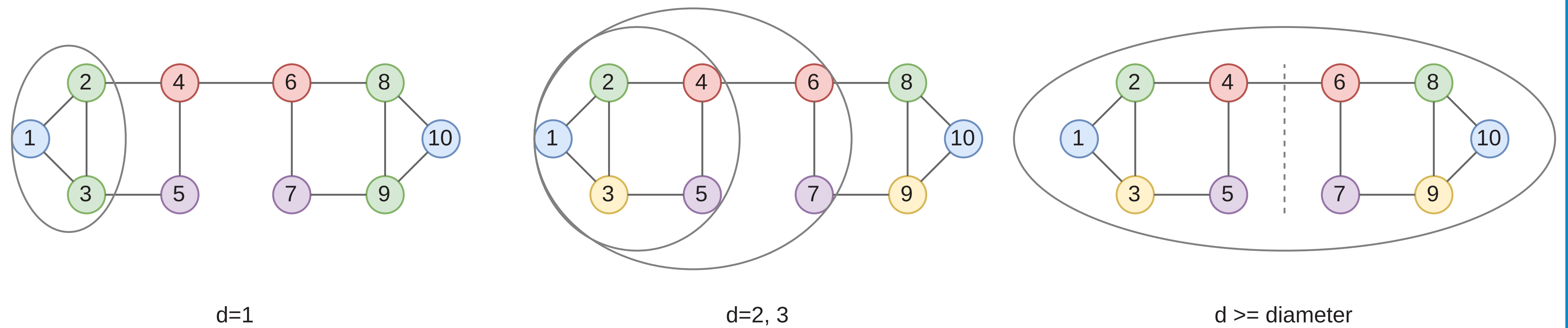


Figure 1: Nodes partitioned using d - k -anonymity, where d -equivalent nodes have the same color.

Anonymity in Graph Models

When using the 1-neighborhood to assess anonymity, results in Figure 2 show that Erdős-Rényi (ER) graphs with **low average degrees** and **large sizes** are **more anonymous**, while **high average degrees** result in a **lower level of anonymity**. This replicates the results of [4]. When using the 2- or 3-neighborhood, the figure changes drastically: graphs are overall much **less anonymous** and increasing the size has less effect.

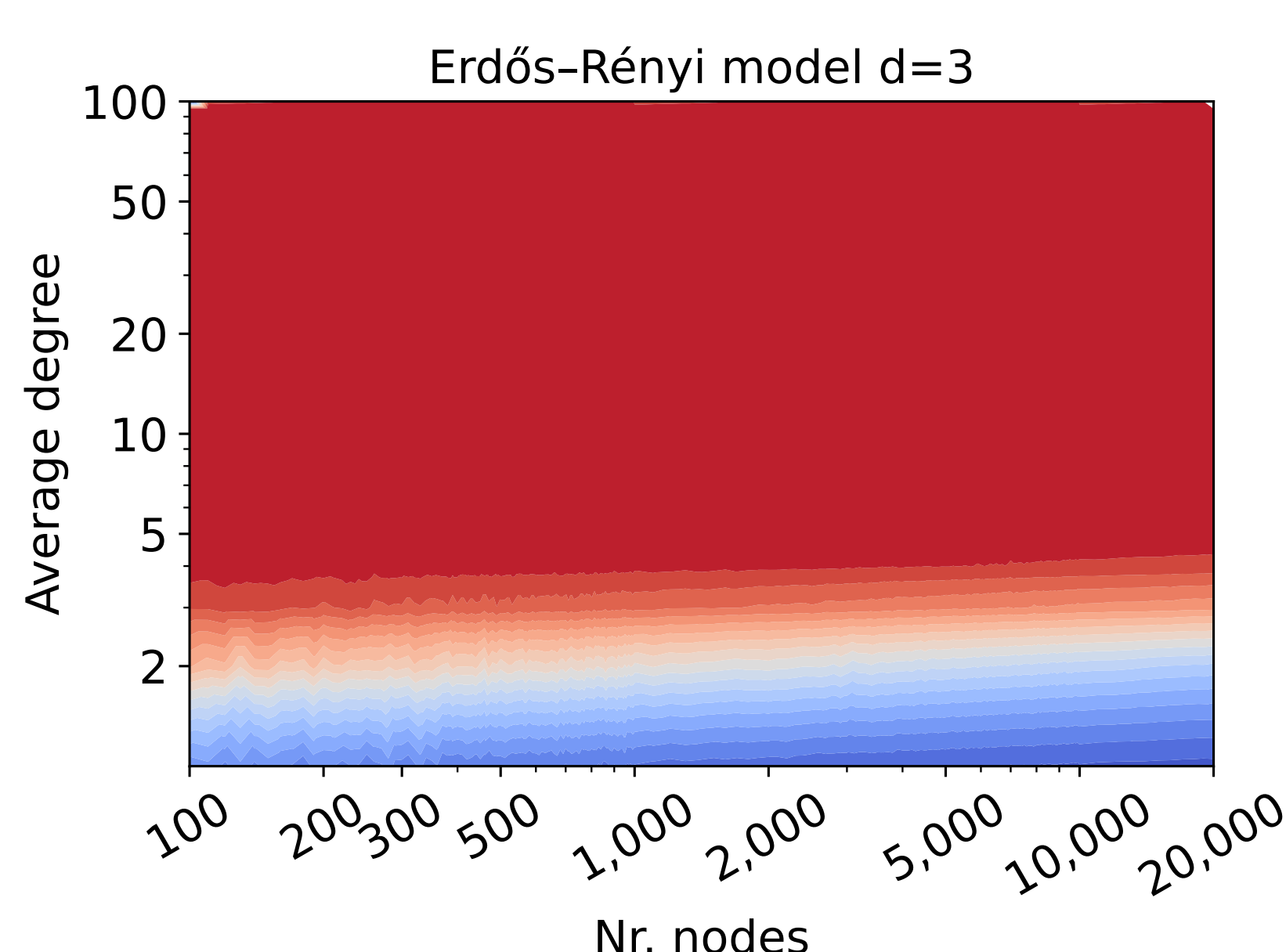
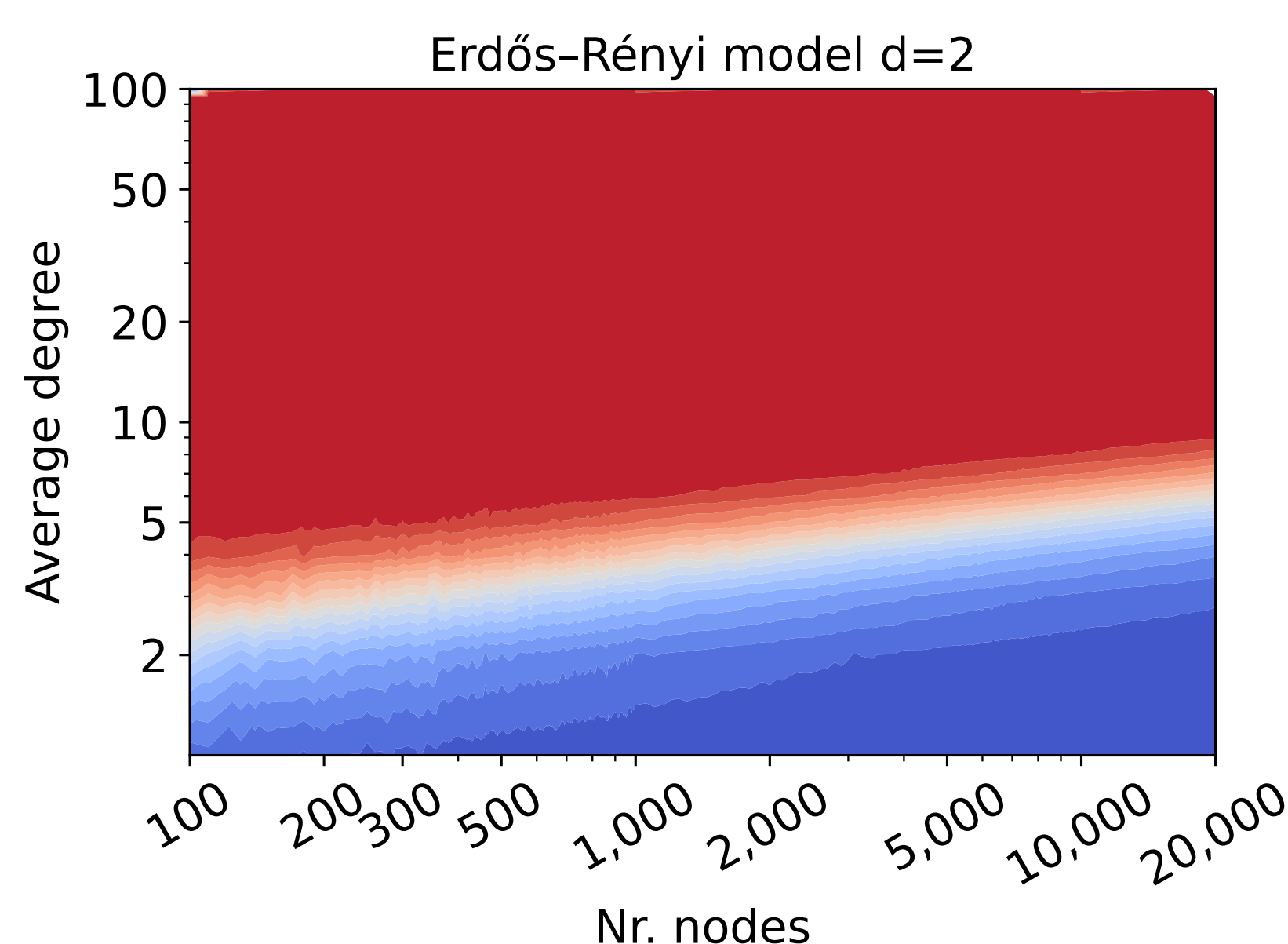
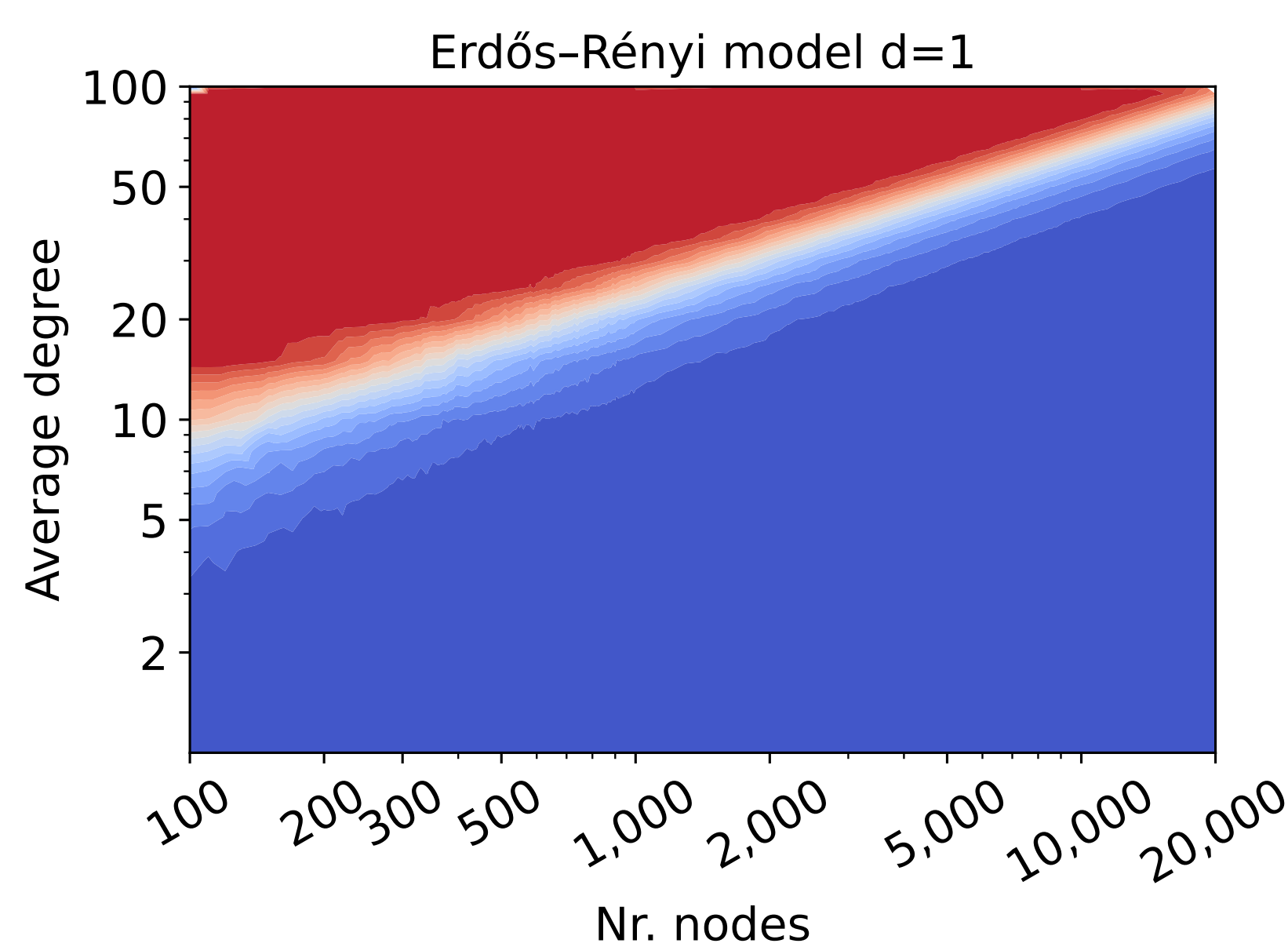


Figure 2: Fraction of unique nodes in ER graphs with various edge densities ($\uparrow$) and sizes ($\rightarrow$).

Empirical Networks

The decrease in anonymity can also be found in empirical networks. Overall, from Figure 3 the following three cases can be distinguished:

- High overall anonymity at larger distances
- Largest decrease in anonymity when $d \geq 2$
- Anonymity is lost when $d = 1$

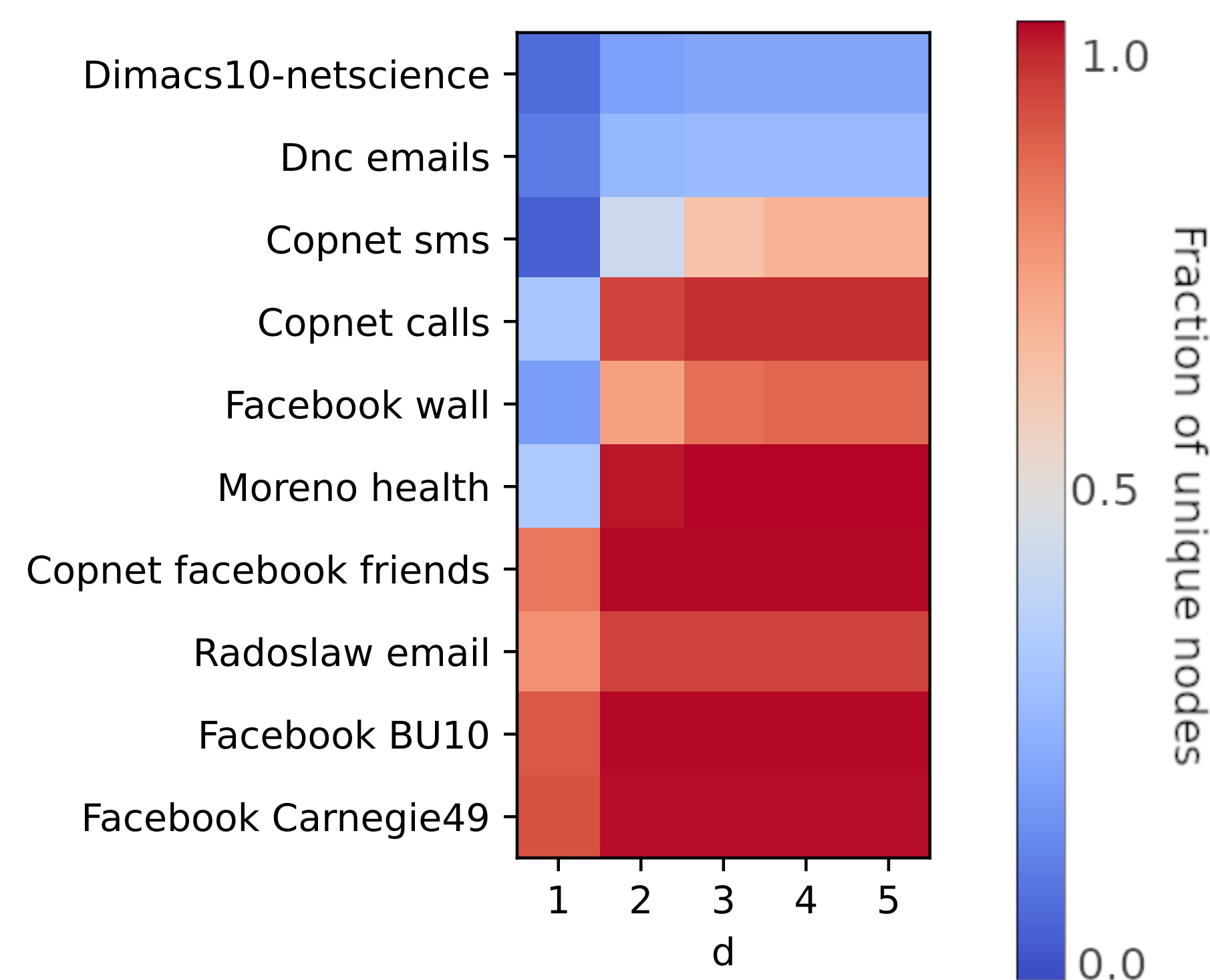


Figure 3: Fraction of unique nodes in empirical networks for different values of d .

References

- [1] Brendan D. McKay and Adolfo Piperno. Practical graph isomorphism, ii. *Journal of Symbolic Computation*, 60:94–112, 2014.
- [2] Rachel G. de Jong. Measuring structural anonymity in complex networks. *MSc thesis, Leiden University*, 2021.
- [3] Rachel G. de Jong, Mark P. J. van der Loo, and Frank W. Takes. Algorithms for efficiently computing exact structural anonymity in complex networks. (*Working paper*).
- [4] Daniele Romanini, Sune Lehmann, and Mikko Kivelä. Privacy and uniqueness of neighborhoods in social networks. *Scientific Reports* 11: 20104, 2021.
- [5] Jan van der Laan and et al. A Whole Population Network and Its Application for the Social Sciences. *European Sociological Review*, 2022.

Acknowledgements & Contact

This research was made possible by the Platform Digital Infrastructure SSH - www.pdi-ssh.nl

Leiden Computational Network Science group - cns.liacs.nl

Questions or comments?
r.g.de.jong@liacs.leidenuniv.nl



ANO-NET

A Population-scale Family Network

Figure 4 shows a part of a family network of parent→child relations where nodes in the five largest equivalence classes are colored. At distance 1 most nodes are in a large (pink) equivalence class and hence anonymous. However, knowledge about multi-hop neighborhoods drastically decreases the anonymity of nodes, even if they have a frequently occurring 1-neighborhood.

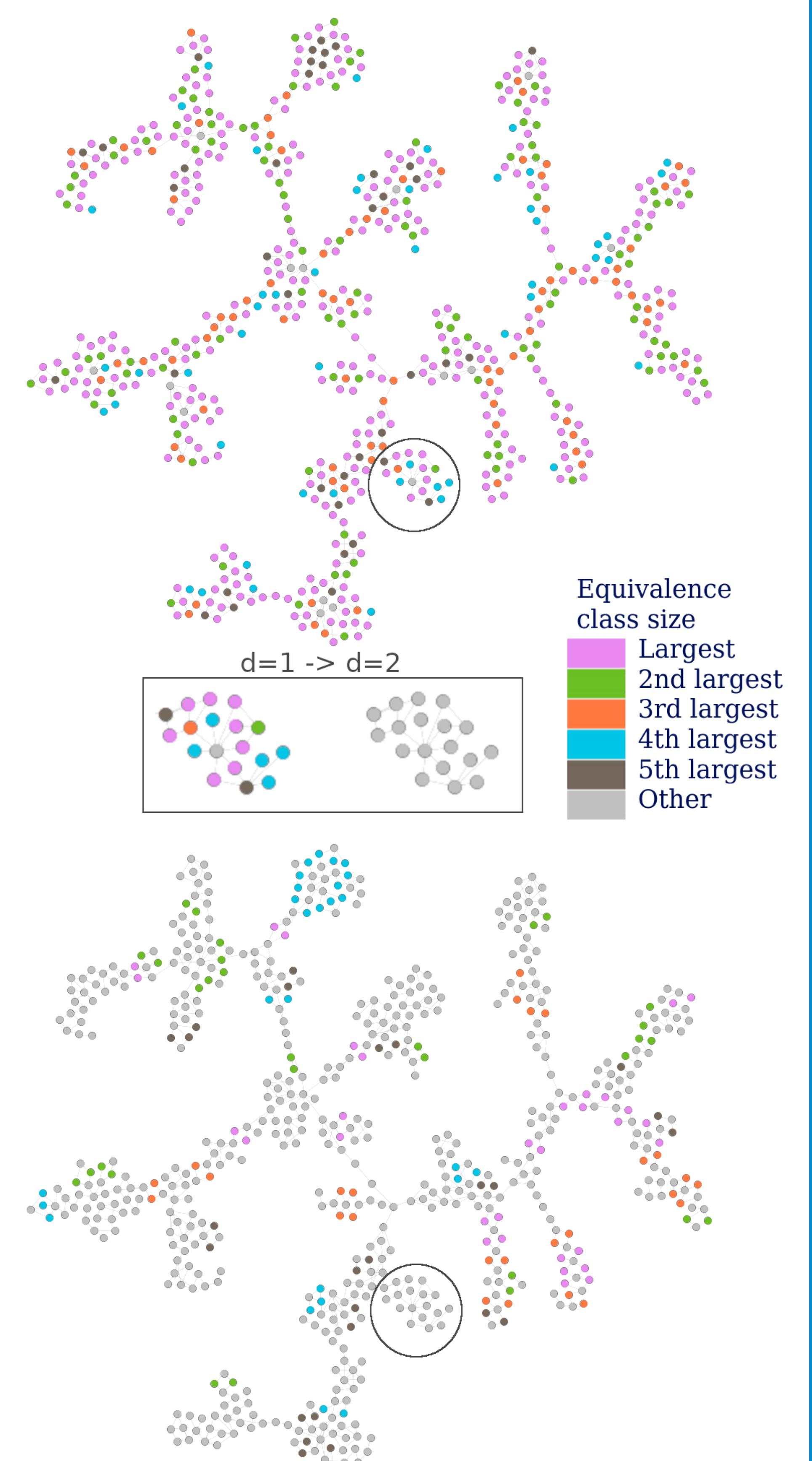


Figure 4: Second largest component of the POPNET family network [5]. Edges represent parent-child relations, 5 largest groups are colored w.r.t. 1- k -anonymity (top), and 2- k -anonymity (bottom).